Advanced Machine Learning (AML-MDS)

Syllabus

Master in Data Science, Fall 2026–27

Lluís A. Belanche

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This is what the course teaches, session by session, in the order it is taught. The dates of every session and every deliverable are in the *course handout*, and only there; so are the assessment, the notation (Part II) and the rules. Sessions are labelled by block: **F** foundations, **K** kernel methods, **N** neural methods, **H** hybrid methods, **E** ensemble methods.

One expression for the whole course. Almost every method in it is an instance of

$$\hat{f}_n = \arg\min_{f \in \mathcal{F}} \; R_n(f) + \lambda\, \Omega(f), \qquad R_n(f) = \frac{1}{n} \sum_{i=1}^{n} \ell\big(y_i, f(\boldsymbol{x}_i)\big),$$

with four slots: the loss ℓ, the penalty Ω (with its weight λ), the way the arg min is computed, and the class $\mathcal{F}$. Block F fills them for the linear model on a fixed basis; the kernel methods stop computing the basis; the neural methods stop fixing it; the hybrid methods bring the two together; and the ensemble methods combine whole fitted models.

Block	Sessions	Laboratories	Project
Foundations	F1–F4	0, 1	proposal
Kernel methods	K1–K5	2, 3, 4 (kernel side)	milestone 1
Neural methods	N1–N3	4 (network side), 5	milestone 2
Hybrid methods	H1	6	milestone 3 (with ensembles)
Ensemble methods	E1–E2	—	milestone 3; final report and oral

1 Foundations

What a perfect answer would look like, why the training error is not it, and the linear model read in three ways. References for the block: Bishop [2], Hastie et al. [11]; for consistency and the theory of ERM, Devroye et al. [7], Vapnik [28]; for generalized linear models, McCullagh and Nelder [15].

F1 — Learning from data
The objects: input and output spaces, the loss, the regularizer, the unknown distribution ρ, the sample D_n, the hypothesis space $\mathcal{F}$; parameters and hyperparameters. Risk and empirical risk (always averaged); conditional risk and pointwise optimality; the Bayes predictor and the Bayes risk. Empirical risk minimisation and the optimism of the training error. Excess risk; the regression function; bias, variance and noise; under- and overfitting. Classification: the Bayes classifier, costs and K classes, the reject option; generative and discriminative strategies. The noise function. Three ways of splitting the risk, and why one of them points to regularization.

F2 — The linear model: statistical view
Least squares from a Gaussian likelihood; the normal equations, algebraically and geometrically; the comet. Basis functions: linear in the parameters, not in the inputs. Computing the fit: the singular value decomposition, and when least squares is unstable. Regularization: ridge,

choosing λ by leave-one-out without refitting and by generalized cross-validation, AIC and BIC, the LASSO. Maximum likelihood in general; the exponential family; generalized linear models, convexity under the canonical link, IRLS; logistic, multinomial, Poisson, binomial and exponential regression; penalised GLMs. Estimators: bias, variance, consistency, the Cramér–Rao bound and the asymptotics of maximum likelihood.

F3 — The linear model: machine learning view
The same model with no distribution and no likelihood. Gradient descent and the step size; batch, mini-batch and stochastic versions. The ADALINE and the delta rule; weight decay and early stopping as shrinkage. Linear classifiers: functional and geometrical margins, the margin of a data set; the perceptron criterion and rule [19]; Novikoff's theorem [17]; the perceptron with margin and the dual form. One linear predictor, four losses.

F4 — Three stories about one model
(for self-study) The linear model read as a GLM and as a network, and where the two readings part. The Bayesian story: the posterior, MAP is ridge, what a posterior gives that a point estimate cannot, and three readings of one λ. Why the linear model is convenient and what it costs: the weak point is the basis. The fork towards kernels and networks.

2 Kernel methods

Do not compute the basis. References for the block: Schölkopf and Smola [21], Shawe-Taylor and Cristianini [25].

K1 — The support vector machine
From the margin of a data set to the widest separating hyperplane [6]; hard and soft margin. The soft margin as the course's expression: hinge loss plus $\lambda\|\boldsymbol{w}\|^2$, with $\lambda = 1/(2nC)$. Lagrangian and Karush–Kuhn–Tucker conditions, where complementary slackness *is* the set of support vectors; the dual problem. VC dimension and the margin bound: margin, not dimension. Leave-one-out bounds. The session ends on the observation that the dual only sees inner products.

K2 — Kernel design
The SVM goes non-linear: feature maps as basis functions, and the kernel trick. Hilbert spaces. Which functions are kernels: positive semi-definite functions; reproducing kernel Hilbert spaces and the reproducing property; the Moore–Aronszajn theorem [1]. Polynomial and Gaussian kernels; closure properties and normalisation. The norm of the space as a regularizer.

K3 — Kernelizing algorithms
The representer theorem [23]: its general and semiparametric forms, existence and uniqueness; kernel ridge regression; the kernel perceptron. Inner products and distances in feature space; conditionally positive semi-definite kernels and the kernel trick for distances; kernel k-nearest neighbours; kernel PCA [22]. Not every kernel method is a regularized fit.

K4 — The SVM, continued
The parameters C and ν; applications (handwritten digits; DNA with categorical kernels). Support vector regression, the course's expression with the ε-insensitive loss [26]; novelty detection with the one-class SVM [24]; practical advice on choosing a kernel. The four slots, filled by the kernel methods.

K5 — Beyond $\mathbb{R}^d$
Kernels for strings, trees and graphs; kernels for categorical, mixed and missing data. Kernel Fisher discriminant analysis [16].

3 Neural methods

Do not fix the basis: learn it. Reference for the block: Goodfellow et al. [10].

N1 — Networks from the foundations
The last layer is a GLM on a learned representation. Composition and depth; universal approximation [12]; backpropagation [20]. A non-convex problem, unlike everything before it.

N2 — Regularization without a canonical norm
Weight decay, the third appearance of $\|\boldsymbol{w}\|_2^2$; dropout [27]; early stopping; normalisation. Autoencoders.

N3 — Variational autoencoders
Latent variables, and the Bayesian machinery of F4 applied: approximate posterior, evidence lower bound, reparametrisation [14]. The Kullback–Leibler term as a regularizer. Diffusion models, announced.

4 Hybrid and ensemble methods

Bring the two families together; then combine whole models.

H1 — Closing the circle
Random Fourier features [18]; the Nyström approximation [29]; neural tangent kernels [13]; deep kernel learning [30].

E1 — Bagging and boosting
Bagging [3] and random forests [4]; AdaBoost [8]; gradient boosting [9] and the XGBoost family [5].

E2 — Stacking
Stacked generalization [31]: to stack or not to stack. Closing remarks.

5 Laboratories and project

Each laboratory is one experiment run by the whole class at once: every pair runs one cell of a grid of configurations, and the pooled result answers a question that no pair can answer alone. From laboratory 4 each pair's cell is its own project dataset. What to install, by when, and how a session works are in the *laboratory instructions*.

Lab	After	Topic	Tool
0	F1–F3	Resampling: estimating the risk of a procedure from the data alone (on the blackboard)	—
1	F1–F4	Foundations: how the best model complexity moves with n and with the noise, and whether cross-validation finds it	R
2	K1–K2	Kernel methods I	R
3	K3–K4	Kernel methods II	R
4	K5, N1	A fixed kernel against a learned representation: SVM, MLP and RBF network	Python
5	N2–N3	Neural methods	Python
6	H1	Hybrid methods	Python

The **project** is one problem carried through the whole course — a dataset and a question about it, or a methodological question studied on synthetic and real data: a proposal, one milestone per family of methods, a final integrative report and an oral defence. It is described in the *project guidelines* and graded with the *rubric*.

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